MEDINOVA TEACHING HOSPITAL

Class 05 — Respiratory Medicine

COPD · Asthma · Pneumonia · Respiratory Failure

> *"Every breath tells a story. Learn to listen."*

JUNIOR RESIDENCY SERIES | VOL. 05

SECTION 01

Respiratory Examination: A Comprehensive Guide

A standard respiratory exam requires a systematic approach: Inspection, Palpation, Percussion, and Auscultation (IPPA). Below are clinical findings for major pathologies.

Condition	Percussion Note	Breath Sounds	Vocal Resonance	Mediastinal Shift
Consolidation (Pneumonia)	Dull	Bronchial (harsh)	Increased	None
Pleural Effusion (Fluid)	Stony Dull	Absent / Diminished	Decreased	Away from lesion (if large)
Pneumothorax (Air)	Hyper-resonant	Absent / Diminished	Decreased	Away (Tension only)
Collapse (Atelectasis)	Dull	Absent / Diminished	Decreased	Toward the lesion
COPD/Asthma	Normal / Hyper-resonant	Vesicular + Wheeze	Normal	None

Definitions for the Junior Resident:

- **Tactile Vocal Fremitus:** Vibration felt on the chest wall when the patient speaks (e.g., "99"). Increased in consolidation because sound travels better through solid than air.
- **Vesicular Breathing:** The soft, rustling sound of normal air entry into healthy lung tissue.
- **Bronchial Breathing:** High-pitched, hollow sounds suggesting air is passing through a patent bronchus into consolidated lung.

> **Prof. Obi's Pearl #1**
> *"The trachea is the 'fingerprint' of the upper mediastinum. If it's not midline, you aren't looking at simple asthma; you're looking at a structural catastrophe."*

SECTION 02

COPD: Chronic Obstructive Pulmonary Disease

COPD is characterized by persistent airflow limitation. Differentiation from Asthma is key.

GOLD Staging (Based on FEV_1 % Predicted)

GOLD 1 (Mild)	FEV1 ≥ 80%	**GOLD 3 (Severe)**	30% ≤ FEV1 < 50%
GOLD 2 (Mod)	50% ≤ FEV1 < 80%	**GOLD 4 (V. Severe)**	FEV1 < 30%

NIV (Non-Invasive Ventilation) Criteria: Used in acute exacerbations with Respiratory Acidosis (pH 7.25 - 7.35) despite optimal medical therapy. If pH < 7.25, consider intubation.

SECTION 03

Asthma: BTS/SIGN Stepladder

1. **SABA:** Short-acting beta-agonist (Salbutamol) PRN.
2. **Low-dose ICS:** Inhaled corticosteroid (Beclometasone) daily.
3. **LTRA:** Leukotriene receptor antagonist (Montelukast) or LABA.
4. **MART:** Maintenance and Reliever Therapy (Combined ICS/LABA).

Severity Classification Table

Moderate	Severe	Life-Threatening
PEFR 50-75% Normal Speech	PEFR 33-50% RR > 25, HR > 110	PEFR < 33% Silent Chest, Cyanosis, Exhaustion

> **Prof. Obi's Pearl #2**
> *"The most dangerous asthma patient is the one who stops wheezing but remains breathless. A 'silent chest' isn't improvement; it's a prelude to respiratory arrest."*

SECTION 04

Community Acquired Pneumonia (CAP)

Management is dictated by the **CURB-65 Score** (1 point each for: Confusion, Urea > 7, RR ≥ 30, BP < 90/60, Age ≥ 65).

Score	Risk	Management Pathway	Antibiotic Choice (Typical)
0 - 1	Low (< 3%)	Home treatment	Oral Amoxicillin (5 days)
2	Moderate (9%)	Hospital admission	Oral/IV Amoxicillin + Clarithromycin
3 - 5	High (15-40%)	Urgent ICU assessment	IV Co-amoxiclav + Clarithromycin

SECTION 05

Respiratory Failure & Oxygen Devices

Type 1: Hypoxaemic (Low O2, Normal/Low CO2). Failure of lung tissue (e.g., Pneumonia, PE).
Type 2: Hypercapnic (Low O2, High CO2). Failure of pump/ventilation (e.g., COPD, Neuromuscular disease).

Oxygen Delivery Comparison

Device	Flow Rate	FiO2 (%)	Clinical Use
Nasal Cannula	1 - 4 L/min	24 - 35%	Non-acute, mild hypoxia.
Venturi Mask	Fixed L/min	24, 28, 35, 40%	COPD (Controlled oxygen).
Non-Rebreather	12 - 15 L/min	80 - 90%	Critical Emergencies/Trauma.

SECTION 06

Pleural Disease Management

Spontaneous Pneumothorax (BTS 2023 Update)

The 2023 British Thoracic Society guidelines emphasize *conservative management* for stable patients, regardless of size.

Primary (No underlying lung disease):
• Is patient symptomatic? If NO → Observation.
• If YES → Aspiration (16-18G). If fails → Chest Drain.

Secondary (Existing lung disease, e.g., COPD):
• > 2cm or symptomatic? → Chest Drain immediately.

Pleural Effusion: Light's Criteria

Used to distinguish **Transudate** (pressure-related) from **Exudate** (inflammation-related). An Exudate meets at least one of:

1. Effusion protein / Serum protein ratio > 0.5
2. Effusion LDH / Serum LDH ratio > 0.6
3. Effusion LDH > 2/3 of upper limit of normal serum LDH.

> **Prof. Obi's Pearl #3**
> *"If Light's Criteria says it's an exudate but the patient looks like they have heart failure, check the protein gap. If it's close to 30g/L, don't rush to biopsy the pleura."*

SECTION 07

Pulmonary Embolism (PE)

PE is the 'Great Mimic'. Use clinical scores to prevent over-testing.

Wells Score for PE

Clinical signs of DVT (3), Alternative dx less likely (3), HR > 100 (1.5), Immobilization/Surgery (1.5), Prev DVT/PE (1.5), Haemoptysis (1), Malignancy (1).

- **Score > 4:** PE Likely. Proceed to CTPA.
- **Score ≤ 4:** PE Unlikely. Perform D-Dimer test.

Anticoagulation Strategies

- **DOACs:** Apixaban or Rivaroxaban (First line for most).
- **LMWH:** Dalteparin (Preferred in pregnancy or malignancy).
- **Thrombolysis:** (e.g. Alteplase) Only for *Massive PE* with hemodynamic instability (BP < 90).

> **Prof. Obi's Pearl #4**
> *"A normal D-Dimer can rule out a PE, but a high D-Dimer cannot rule it in. It's a test of exclusion, not confirmation."*

OFFICIAL SIGN-OFF

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SERIES: Junior Residency Module 05 of 20
DATE: March 2026